Rheumatic heart disease secondary prevention : An investment-case model for country programs

Expert meeting discussion slides

October 2026

Overview

- Study objectives
- Model design/schematic
- Table of data inputs
- Description of analysis plan/scenarios
- Preliminary results (updated abstract needed)
- Questions for experts

Study Objectives

- To estimate health outcomes of scaling up secondary prevention services (RHD screening and secondary antibiotic prophylaxis (SAP)
- To estimate the cost-effectiveness and budget impact of scaling up RHD screening and SAP
- To develop an online tool for country programs to assess value for money in RHD screening based on the experience with ADUNU [An expansion of ADUNU Aim 2].

This study aims to answer the question;

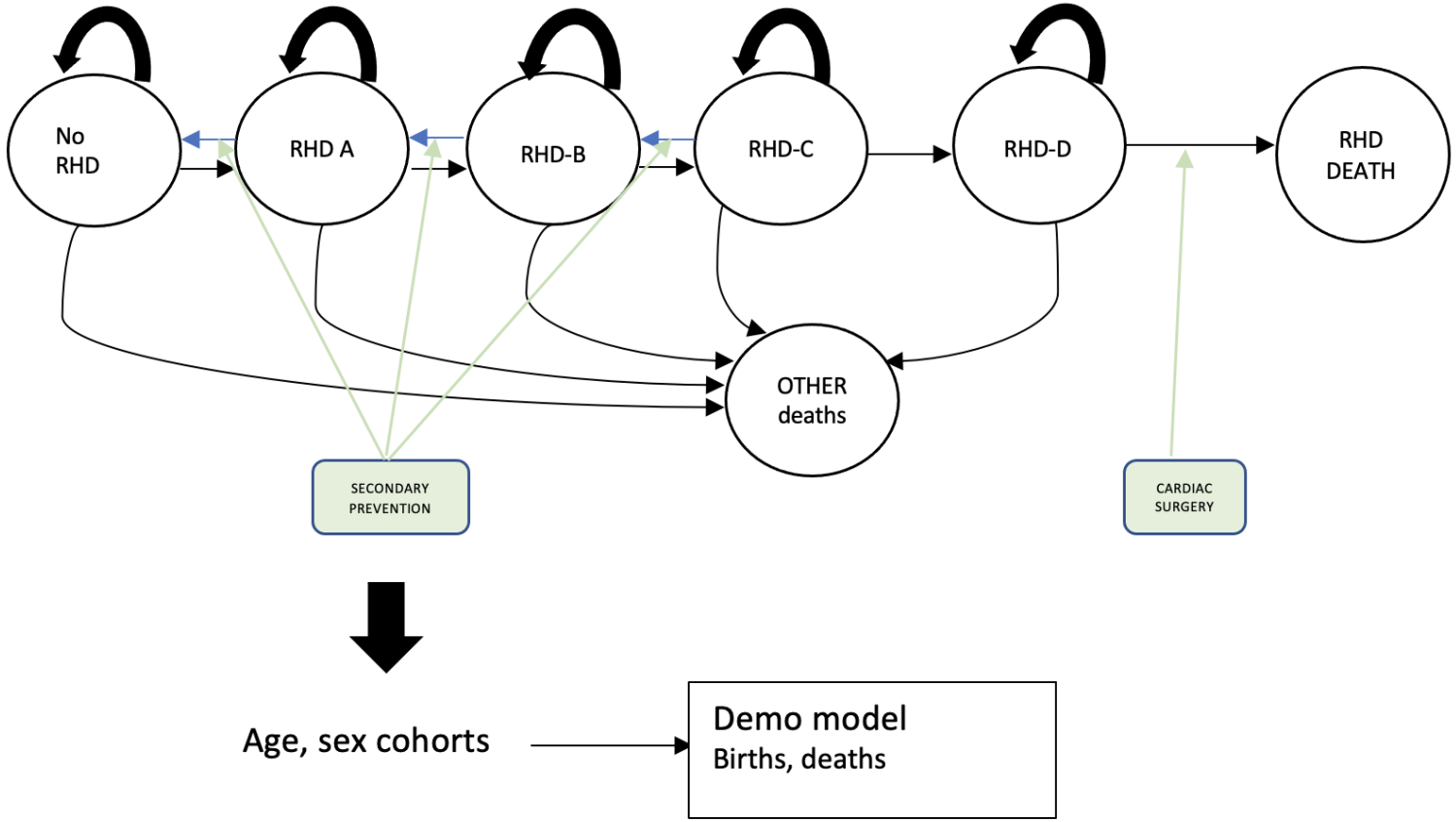
How many RHD deaths does scaling up echocardiography screening and secondary antibiotic prophylaxis avert in two countries (Uganda and Indonesia) over the analysis horizon (2026–2050; 2026 - 2100), and does the monetised health benefits justify the incremental cost?

Model design/schematic

Cohort state transition model, matrix-form (**age** x **sex**) Markov cohort model
Structured after Coates et al., Lancet Global Health 2021 (the global RHD investment-case framework)

Incorporates the **2023 World heart Federation** stages A (minimal), B (mild established), C (advanced without complications), D (advanced with complications)

Uses GBD 2023 RHD epidemiology data (prevalence, mortality), adapted to the demography of Uganda and Indonesia respectively



Model design/schematic

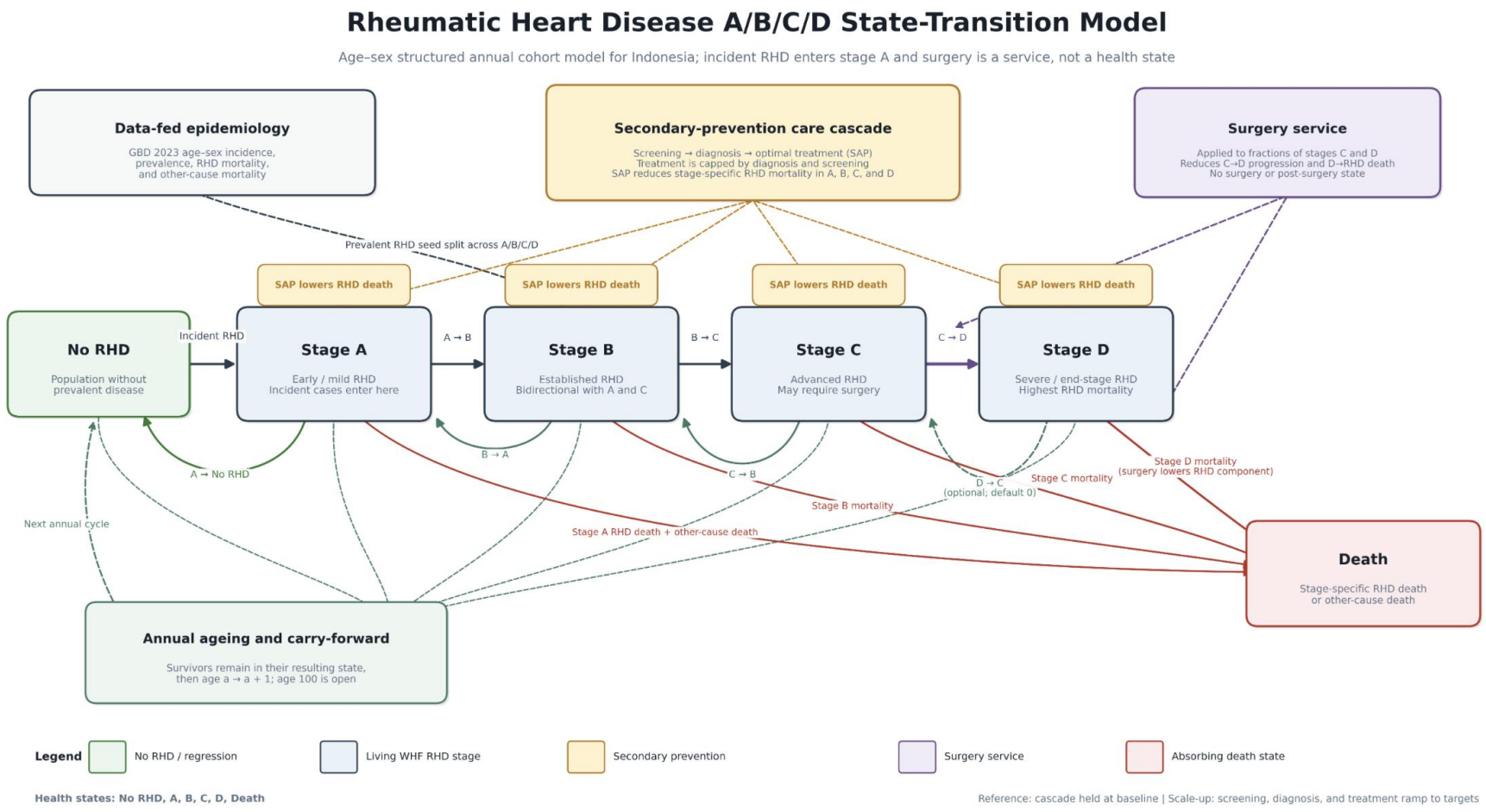


Figure M1. Rheumatic heart disease A/B/C/D model structure. An age–sex structured annual cohort for Indonesia. Incident RHD enters at stage A; the four living stages progress and regress along A ↔ B ↔ C with forward progression C → D; stage D feeds RHD death; competing background (non-RHD) mortality acts on every living stage. Secondary antibiotic prophylaxis (SAP) reduces stage-specific RHD mortality; surgery is a clinical service that reduces C → D progression and D → RHD death (it does not remove patients from a stage). Survivors age one year per cycle, with an open terminal 100+ group. Acute rheumatic fever is not represented as a health state.

Description of analysis plan/scenarios (add more detail)

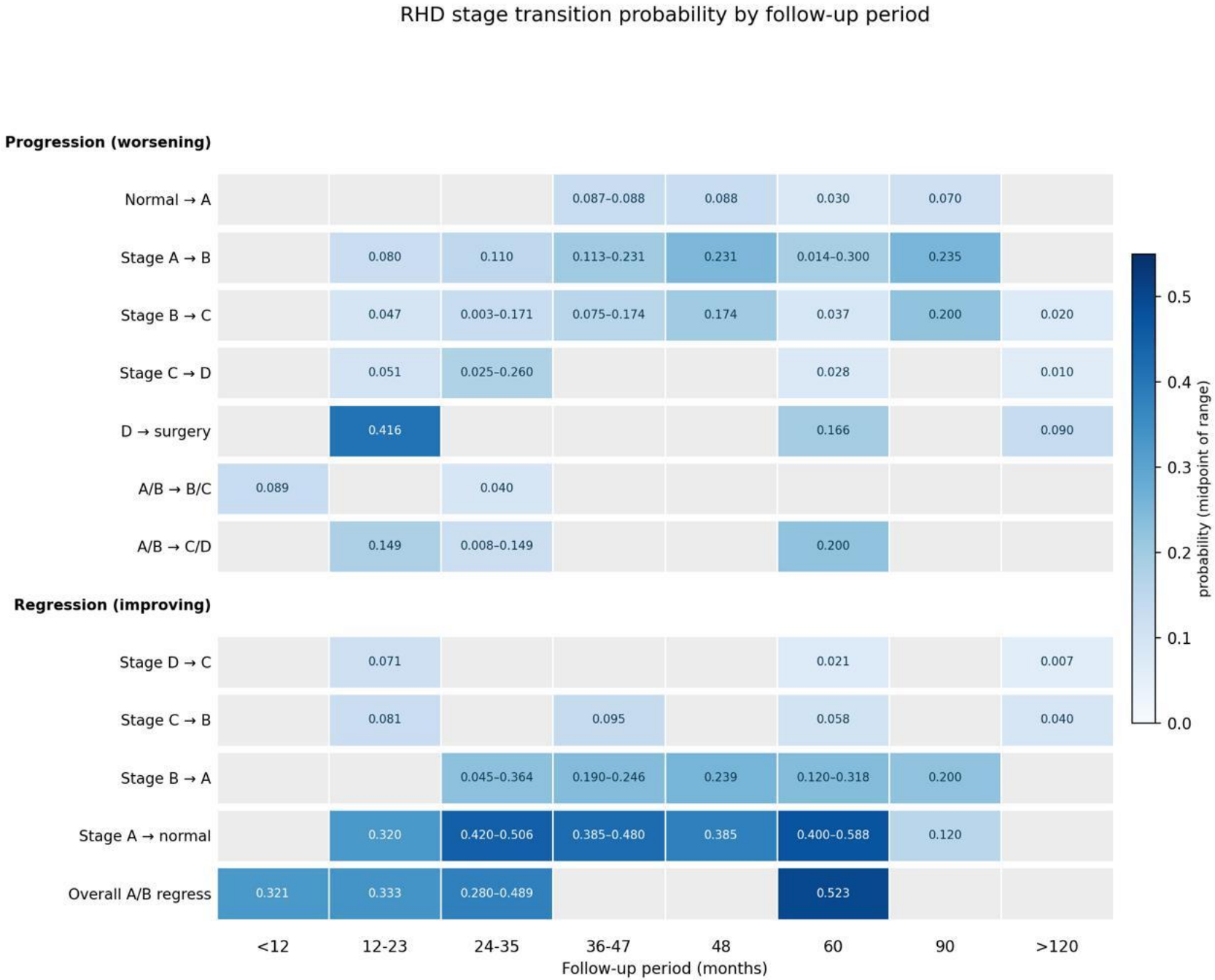
- To estimate impact, we scale up interventions from current coverages to the targets:
  - 80% for screening and diagnosis
  - 65% optimal SAP treatment

- Two arms;
  - a reference arm, with the care cascade held at its baseline coverage, and
  - a SAP scale-up arm, in which screening, diagnosis and optimal treatment rise to their targets

Literature review – transition probabilities

Model Input Parameters

Transition Probabilities				
Transition	**Rates (varying follow up times)**	**Annualized rates**	**Calibrated Sample model (Uganda)**	**Calibrated Sample model (Indonesia)**
No RHD – A*	0.03 – 0.11	0.001 – 0.057		
A – No RHD (Regression)	0.12 – 0.59	0.007 – 0.249	17.8%	7.1%
A – B (Progression)	0.003 – 0.235	0.002 – 0.072	0.8%	1.1%
B – A (Regression)	0.05 – 0.36	0.004 – 0.203	0.6%	1.6%
B – C (Progression)	0.02 – 0.26	0.001 – 0.125	1.3%	4.4%
C – B (Regression)	0.04 – 0.1	0.012 – 0.081	0.5%	0.5%
C – D (Progression)	0.01 – 0.48	0.006 – 0.250	3.6%	8.2%
D – C (Regression)	0.007 – 0.07	0.004 – 0.007	0%	0%
(A/B – B/C overall)	0.04 - 0.09	0.020 – 0.170	-	-
(A/B – C/D overall)	0.08 – 0.149	0.04 – 0.149	-	-
(A/B regress overall)	0.280 – 0.523	0.138 – 0.539	-	-


RHD stage transition probability by follow-up period
Progression (worsening)
Normal → A
Stage A → B
Stage B → C
Stage C → D
D → surgery
A/B → B/C
A/B → C/D
Regression (improving)
Stage D → C
Stage C → B
Stage B → A
Stage A → normal
Overall A/B regress
0.087–0.088
0.088
0.030
0.070
0.080
0.110
0.113–0.231
0.231
0.014–0.300
0.235
0.047
0.003–0.171
0.075–0.174
0.174
0.037
0.200
0.020
0.051
0.025–0.260
0.028
0.010
0.416
0.166
0.090
0.089
0.040
0.149
0.008–0.149
0.200
0.071
0.021
0.007
0.081
0.095
0.058
0.040
0.045–0.364
0.190–0.246
0.239
0.120–0.318
0.200
0.320
0.420–0.506
0.385–0.480
0.385
0.400–0.588
0.120
0.321
0.333
0.280–0.489
0.523
<12
12-23
24-35
36-47
48
60
90
>120
Follow-up period (months)
probability (midpoint of range)
0.0
0.1
0.2
0.3
0.4
0.5

Transition probabilities

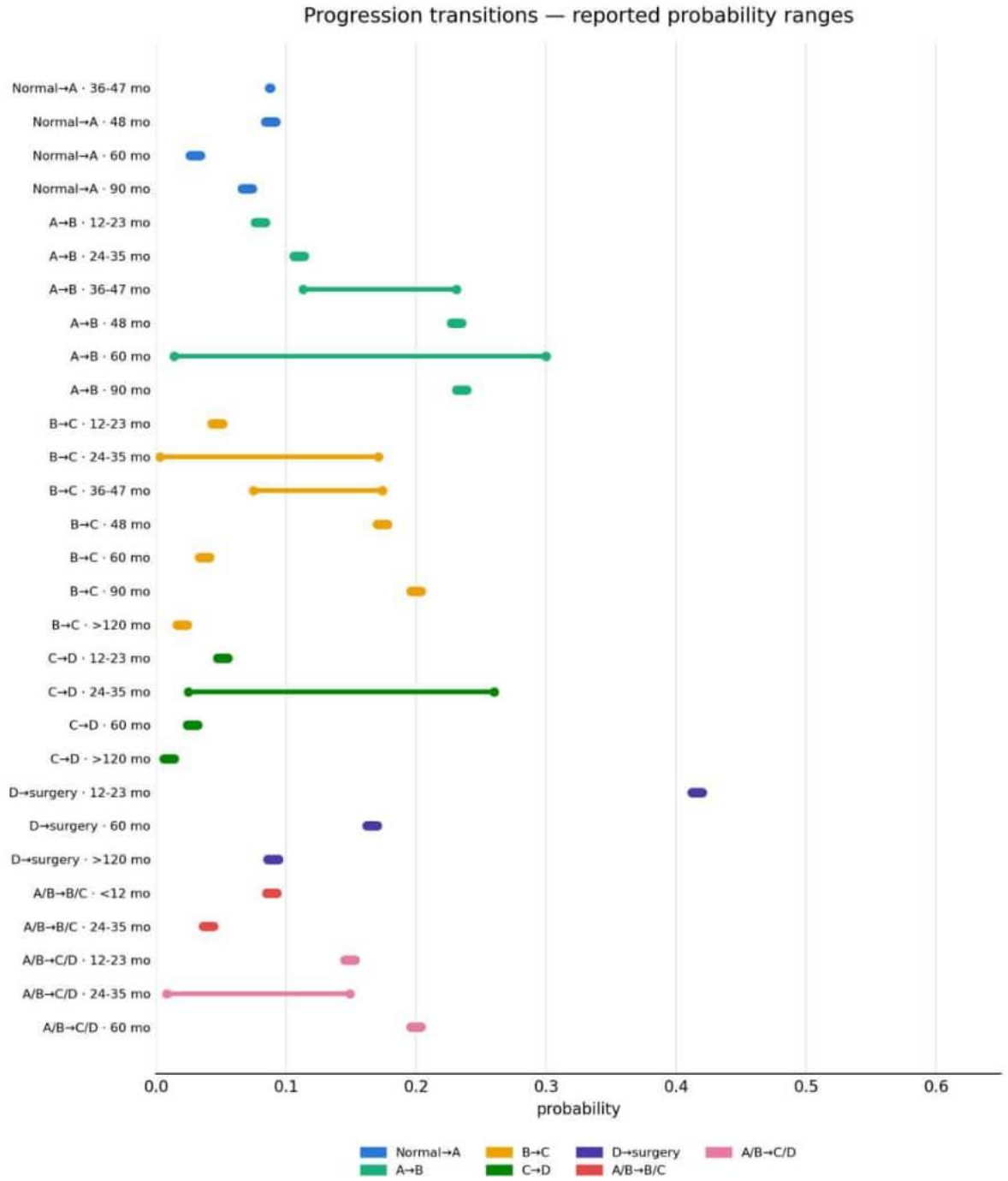


Table of data inputs

Stage RHD specific mortality				
Stage A			0.07%	0.13%
Stage B			0.39%	0.57%
Stage C	3.8% (Cannon)		1.64%	2.26%
Stage D	8%? (REMEDY)		6.63%	9.06%
Intervention effects				
SAP on RHD mortality		55%		
Surgery on C – D		85%		
Surgery on D – RHD death		85%		
OpRHD – dead	3% (Cannon; 6.7% Moireira et al)			
Surgery service**				
Fraction of stage C requiring surgery			10%	3.0%
Fraction of stage D requiring surgery	30 – 40 % (Cannon, severe RHD)		40%	20.0%
Surgery coverage			3%	5%
Stage split (Prevalent pool) (Cannon Table 2**)				
Stage A	56.5%		85.4%	81.1%
Stage B	27.2%		12.1%	11.7%
Stage C	16.2%		1.8%	4.2%
Stage D	0.9%(Ug) vs. 3.2% (Indonesia) (GBD with HF fraction) Discuss – might be closer to 5% of RHD Cases)		0.7%	3.0%

*From studies following controls; **Data from patients aged 5 – 24 years diagnosed between 1999 – 2012 in Australia

Preliminary results (WCRHD abstract, to update)

Development of a tool to conduct investment cases for RHD screening and secondary prevention programs.

Background
The 2024 WHO guidelines for rheumatic heart disease (RHD) included endorsement of echocardiography-based screening. However, because of limited resources, additional evidence is needed to convince policymakers of the value of RHD programs. We present a prototype mathematical modeling tool to conduct "investment cases" (benefit-cost analyses) of policies to expand access to screening and secondary prevention.
Methods:
Our tool integrated three components: (1) a cohort state-transition model (representing RHD stages A-D), (2) a demographic projection model, and (3) an intervention costing model. The tool assessed the cost and impact of scaling up the two interventions on the RHD "cascade of care" and subsequent RHD cases and deaths over 2026 to 2050. The tool was parameterized using Global Burden of Disease and UN Population estimates and intervention cost and effect sizes from the literature. We illustrate the tool's use for a hypothetical low-income African country (2% prevalence, 50 million population) and upper-middle-income Asian country (0.5% prevalence, 300 million population). For both countries, we modeled program scale-up from low current coverage levels to targets of 80% screening coverage and 65% optimal adherence to secondary antibiotic prophylaxis for RHD by 2050.
Results:
For the hypothetical low- and upper-middle-income country, respectively, achieving these targets could avert 7,900 and 23,000 RHD deaths by 2050, representing a 30% and 24% reduction in RHD death rates in 2050. The additional cost would be $0.3 and $0.3 per capita on average annually, representing an additional 0.02% and 0.006% of current GDP per capita. The net economic benefit, discounted at 3%, would be $91 million and $1.1 billion, with benefit-cost ratios of 1.28 and 1.42. The default parameter values in the tool can be readily replaced with values from other settings to generate results for specific countries or subnational units.
Conclusion:
RHD screening programs appear to be cost-beneficial in countries at varying levels of income and endemicity, though with considerable variation. Our tool provides a structured approach for using local data (e.g., prevalence studies, pilot program costs) to generate results that can be used for program planning and policy advocacy.

Keywords: RHD programs, Secondary prevention, Implementation tools, Investment case, Costing, Policy, RHD burden

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Add result figures?

Questions

1. Confirm which stages SAP proven to work: does SAP reduce RHD mortality at stage D - C? (Here we take C to D as only forward progression)

2. Discuss transitions A - D estimates ((Refer to the figure on literature progression rates slides 7/8)

3. RHD specific mortality occurs in which stages? Also discuss severe/fulminant ARF carditis here

4. Surgery for both stage C and D?

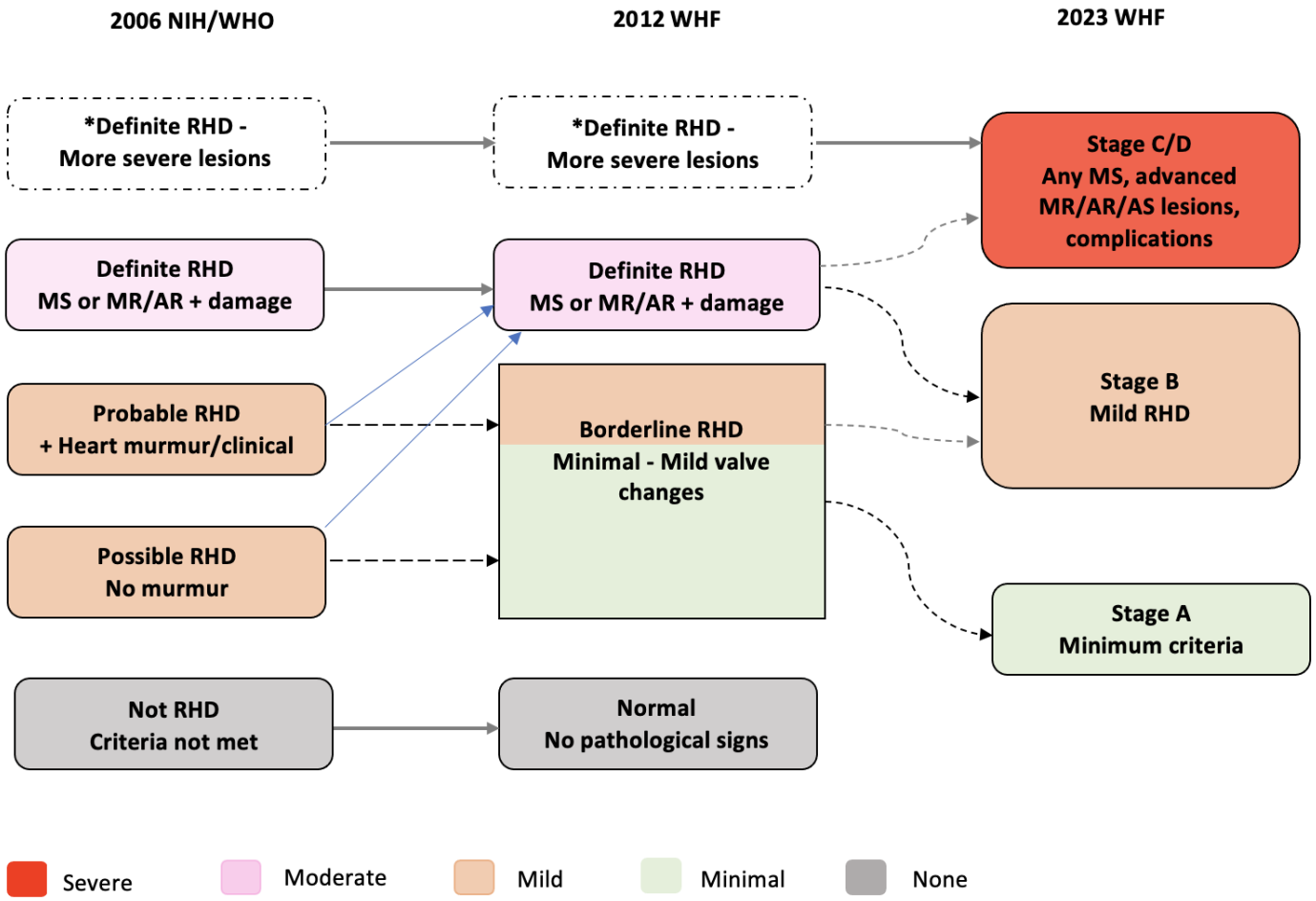
Questions

5. Exact surgery effect: how does surgery impact the pathway: i.e. conceptually more preventing D to death or move to an earlier stage or?

6. Fraction of each stage requiring surgery (literature)

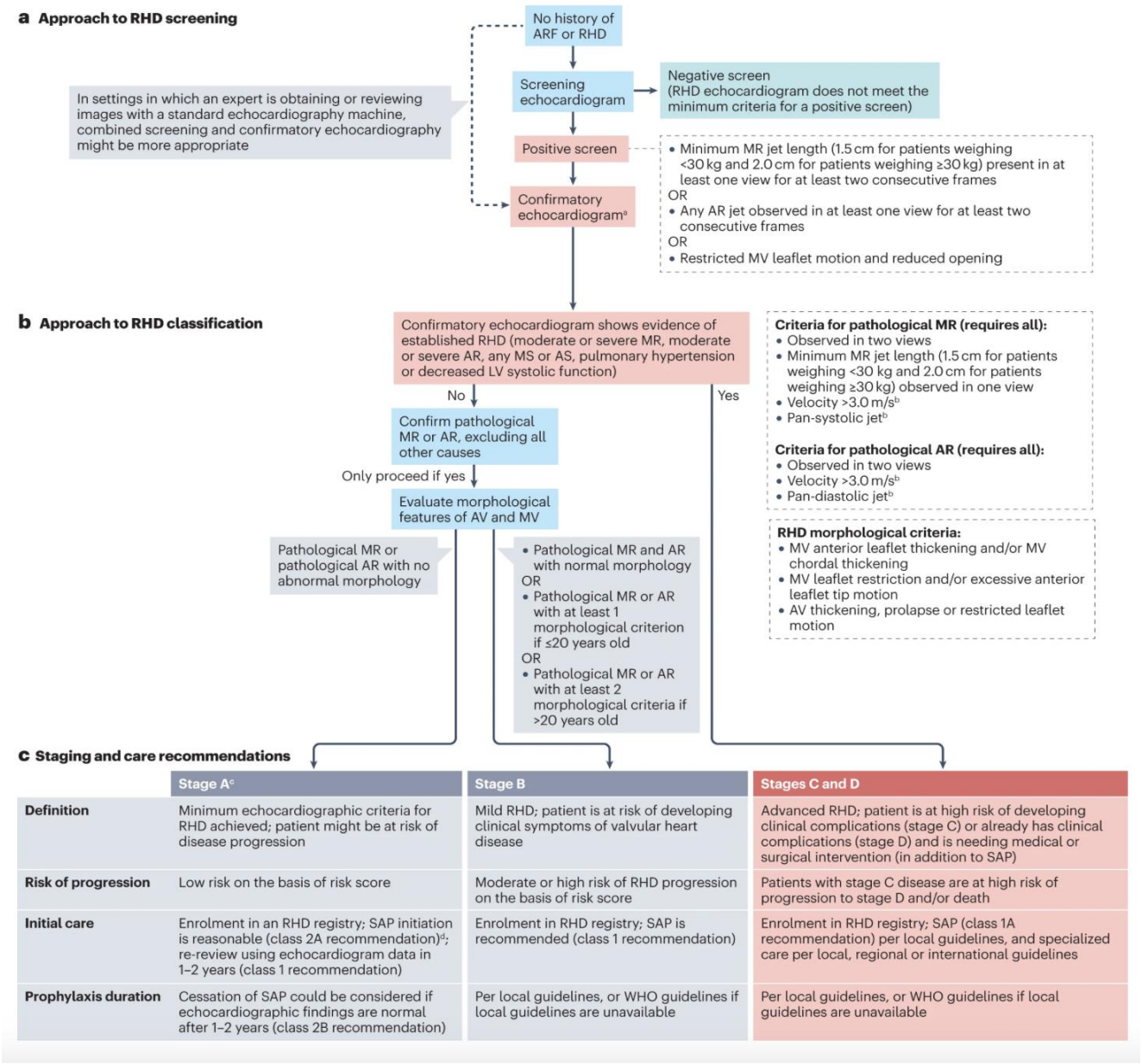
7. Approximate estimates for share of stage A to D in the population/literature (Slide 10)

Extra slides - RHD Criteria matched overtime



*These are added for illustrative purposes

Extra slides – WHF 2023 RHD Guideline



	Stage A[c]	Stage B	Stages C and D
Definition	Minimum echocardiographic criteria for RHD achieved; patient might be at risk of disease progression	Mild RHD; patient is at risk of developing clinical symptoms of valvular heart disease	Advanced RHD; patient is at high risk of developing clinical complications (stage C) or already has clinical complications (stage D) and is needing medical or surgical intervention (in addition to SAP)
Risk of progression	Low risk on the basis of risk score	Moderate or high risk of RHD progression on the basis of risk score	Patients with stage C disease are at high risk of progression to stage D and/or death
Initial care	Enrolment in an RHD registry; SAP initiation is reasonable (class 2A recommendation)[d]; re-review using echocardiogram data in 1–2 years (class 1 recommendation)	Enrolment in RHD registry; SAP is recommended (class 1 recommendation)	Enrolment in RHD registry; SAP (class 1A recommendation) per local guidelines, and specialized care per local, regional or international guidelines
Prophylaxis duration	Cessation of SAP could be considered if echocardiographic findings are normal after 1–2 years (class 2B recommendation)	Per local guidelines, or WHO guidelines if local guidelines are unavailable	Per local guidelines, or WHO guidelines if local guidelines are unavailable

a, Screening for rheumatic heart disease (RHD). [a]Although confirmatory echocardiography is recommended before the initiation of secondary antibiotic prophylaxis (SAP), in settings in which confirmatory echocardiography cannot be performed immediately, SAP can be initiated in patients while waiting for a confirmatory study. The final decision should be made by or in consultation with a clinician with expertise in RHD diagnosis. **b**, Classification of RHD. [b]These criteria can be achieved in two separate views or from two separate continuous-wave Doppler measures. **c**, Staging and care recommendations. [c]Stage A only applies to individuals aged 20 years or less. [d]Evidence from a randomized, controlled trial showed that SAP reduces the risk of RHD progression in children aged 5–17 years with stage A or stage B RHD[4]. However, SAP is also associated with adverse effects, including allergy and anaphylaxis, and numerous health-system factors should be considered before the roll-out of large-scale SAP programmes. Consequently, there is a need to consider family and clinician decisions, as well as health-system factors before recommending SAP in individuals aged ≤ 20 years with stage A RHD who are at low risk of RHD progression. However, all children with stage A disease must have access to follow-up echocardiography and longitudinal clinical evaluation to monitor for disease progression. AR, aortic regurgitation; ARF, acute rheumatic fever; AS, aortic stenosis; AV, aortic valve; LV, left ventricular; MR, mitral regurgitation; MS, mitral stenosis; MV, mitral valve.

Extra slide - WHF 2012 RHD Guidelines

Box 1 | 2012 WHF criteria for echocardiographic diagnosis of RHD

Echocardiographic criteria for individuals aged ≤20 years

Definite RHD (either A, B, C, or D):

- A) Pathological MR and at least two morphological features of RHD of the MV
- B) MS mean gradient ≥4 mmHg*
- C) Pathological AR and at least two morphological features of RHD of the AV‡
- D) Borderline disease of both the AV and MV§

Borderline RHD (either A, B, or C):

- A) At least two morphological features of RHD of the MV without pathological MR or MS
- B) Pathological MR
- C) Pathological AR

Normal echocardiographic findings (all of A, B, C, and D):

- A) MR that does not meet all four Doppler echocardiographic criteria (physiological MR)
- B) AR that does not meet all four Doppler echocardiographic criteria (physiological AR)
- C) An isolated morphological feature of RHD of the MV (for example, valvular thickening) without any associated pathological stenosis or regurgitation
- D) Morphological feature of RHD of the AV (for example, valvular thickening) without any associated pathological stenosis or regurgitation

Echocardiographic criteria for individuals aged >20 years

Definite RHD (either A, B, C, or D):

- A) Pathological MR and at least two morphological features of RHD of the MV
- B) MS mean gradient ≥4 mmHg*
- C) Pathological AR and at least two morphological features of RHD of the AV, only in individuals aged <35 years‡
- D) Pathological AR and at least two morphological features of RHD of the MV

*Congenital MV anomalies must be excluded. Furthermore, inflow obstruction due to nonrheumatic mitral annular calcification must be excluded in adults. ‡Bicuspid AV, dilated aortic root, and hypertension must be excluded. §Combined AR and MR in high prevalence regions and in the absence of congenital heart disease is regarded as rheumatic.
Abbreviations: AR, aortic regurgitation; AV, aortic valve; MR, mitral regurgitation; MS, mitral stenosis; MV, mitral valve; RHD, rheumatic heart disease; WHF, World Heart Federation.

Extra slides – Australian 2012 guidelines

Recommended routine review and management plan for ARF and RHD

Classification	Criteria*	Review and management plan	Frequency†
Priority 1 (severe)¥	Severe valvular disease **or** moderate/severe valvular lesion with symptoms **or** mechanical prosthetic valves, tissue prosthetic valves and valve repairs, including balloon valvuloplasty	Secondary prophylaxis (BPG)	3–4 weekly
		Doctor review	3–6 monthly
		Cardiologist/physician/paediatrician review	3–6 monthly
		Influenza vaccination	Yearly
		Echocardiography	3–6 monthly
		Dental review	Within 3 months of diagnosis, then 6 monthly thereafter
		Pneumococcal vaccination	Refer to *Immunisation handbook*
		Endocarditis prophylaxis	As required Refer to *Therapeutic Guidelines: Antibiotics 2010*
Priority 2 (moderate)	Any moderate valve lesion in the absence of symptoms, and with normal left ventricular function	Secondary prophylaxis (BPG)	4-weekly
		Doctor review	6-monthly
		Influenza vaccination	Yearly
		ECG (optional)	Yearly
		Cardiologist/physician/paediatrician review	Yearly
		Echocardiography	Yearly
		Dental review	Within 3 months of diagnosis, then 6 monthly
		Pneumococcal vaccination	Refer to *Immunisation handbook*
		Endocarditis prophylaxis	As required Refer to *Therapeutic Guidelines: Antibiotics 2010*
Priority 3 (mild)	ARF with no evidence of RHD **or** trivial to mild valvular disease	Secondary prophylaxis (BPG)	4 weekly
		Doctor review	Yearly
		Echocardiography	Children: 2 yearly‡ Adults: 2–3 yearly‡
		Dental review	Yearly

Extra slides– NIH/WHO 2006 Criteria

Diagnostic criteria for rheumatic heart disease in children

Definite Rheumatic Heart Disease (either A or B):

A. Significant mitral stenosis, defined as echocardiographic evidence of mitral stenosis with a mean diastolic pressure gradient across the mitral valve of greater than 4 mm and clinical findings of mitral stenosis with or without other valvular lesions. Such abnormalities as dog-leg deformity of the anterior mitral valve leaflet, fixed or restricted mitral leaflet abnormality, calcification, and commissural thickening were expected.

B. The presence of a heart murmur consistent with any combination of mitral regurgitation or aortic regurgitation and echo-Doppler evidence of rheumatic valvular damage, defined as any of:

(1) significant mitral regurgitation, with a mitral regurgitation jet of at least 2 cm from the coaptation point of the valve leaflets, seen in 2 planes, high velocity (mosaic pattern), holosystolic, plus thickened mitral valve leaflets and/or dog-leg deformity of the anterior mitral leaflet. Additional changes might include multiple regurgitant jets, especially posterolaterally directed.

(2) Significant aortic regurgitation, defined as an aortic regurgitation jet at least 1 cm from the coaptation point of the valve leaflets, of high velocity, seen in 2 planes, plus thickened mitral leaflets and/or dog-leg deformity, without another apparent etiology for the aortic insufficiency, such as a bicuspid aortic valve or annuloaortic ectasia. Aortic stenosis might be associated, but aortic stenosis without associated rheumatic mitral valve disease was not accepted as evidence of rheumatic valvular disease.

Probable Rheumatic Heart Disease

The presence of a heart murmur consistent with any combination of mitral regurgitation or aortic regurgitation, and the subject comes from a population with known or suspected high rates of acute rheumatic fever and/or rheumatic heart disease and no past history of definite or probable acute rheumatic fever, and any of the following findings are present on echocardiography:

(1) Thickened mitral valve leaflets and/or dog-leg deformity of the anterior mitral valve leaflet without significant mitral stenosis.

(2) Significant mitral regurgitation, as defined under Definite rheumatic heart disease, without thickened mitral valve leaflets and/or dog-leg deformity of the anterior mitral valve leaflet.

(3) Significant aortic regurgitation, as defined under Definite rheumatic heart disease, without thickened mitral valve leaflets and/or dog-leg deformity of the anterior mitral valve leaflet.

Possible Rheumatic Heart Disease

The absence of a valvular heart murmur in a subject from a population with known or suspected high rates of acute rheumatic fever and/or rheumatic heart disease with any of the following echo-Doppler changes:

(1) Thickened mitral valve leaflets and/or dog-leg deformity of the anterior mitral valve leaflet.

(2) Significant mitral regurgitation, as defined under Definite rheumatic heart disease.

(3) Significant aortic regurgitation, as defined under Definite rheumatic heart disease.